

A Multi-Model AI Framework for Non-Invasive Stress Detection Using Multimodal Behavioural and Facial Indicators

Sahil Soni¹, Zulfikar Ali Ansari^{1,*}, Wasim Khan¹, K.Kiran Kumar², Shreya Jagtap¹, and Vishal Shah¹

¹AIML Department, Symbiosis Institute of Technology, Symbiosis International (Deemed University), Lavale, Pune, Maharashtra-412115, India

²Department of Computer Science and Engineering, Koneru Lakshmaiah Education Foundation Vaddeswaram, Guntur, Andhra Pradesh, India.

*zulfikar.ansari@sitpune.edu.in

ABSTRACT

In the latest working strategy, workplace stress is a significant issue that impacts workers' health, productivity, and mental well-being over time. Wearable sensors, self-report surveys, and IoT-based physiological systems are among the current methods for tracking stress. However, they often encounter problems, such as high setup costs, user discomfort, concerns about data privacy, and a limited ability to detect real-time changes in stress. To handle these challenges, our study introduces Stress-Scape, a multimodal, software-only artificial intelligence framework that combines facial expression analysis with behavioural indicators derived from keystroke dynamics. The proposed system operates solely on standard computing devices, enabling non-intrusive, real-time stress assessment. We have utilised the facial component with three transfer learning architectures, such as MobileNetV2, EfficientNetB0, and ResNet50V2, trained on 53,571 augmented images from the JAFFE dataset, categorised into binary stress classes. ResNet50V2 achieved the best results, with 89% accuracy, 0.95 precision, and an AUC-ROC of 0.83. The behavioural component uses features such as error rate, backspace frequency, and typing rhythm, modelled with a Random Forest (RF) classifier trained on the CMU Keystroke Dynamics Benchmark dataset. With an optimised decision threshold ($\gamma = 0.65$), this model attains 84.56% accuracy, 0.92 precision, and 0.8567 recall across group-aware cross-validation folds. Combining both modalities provides a more stable and reliable measure of occupational stress, with a final fused confidence of 93.7% during real-time deployment at 25 to 30 FPS. The system features local inference, temporal smoothing, and privacy-preserving processing, enabling practical continuous monitoring without the need for physiological sensors. Finally, our study demonstrates that ambient behavioural cues and facial micro-expressions can serve as effective computational proxies for stress, offering a scalable, non-invasive approach for workplace wellness, cognitive load assessment, and future human-computer interaction systems.

Keywords: Stress Detection, Multimodal AI, Transfer Learning, Keystroke Dynamics, Facial Expression Recognition, Random Forest, CNN, ResNetV2, Affective Computing, Digital Well-being, Workplace Wellness

Introduction

Nowadays, stress is a major concern in our human life as working employees. It makes people less effective, less mentally healthy, and puts more strain on organisations¹. Approximations from international health agencies suggest that over 80% of employees face ongoing workplace stress, leading to substantial economic and public health impacts². With the quick shift to remote and hybrid work arrangements, stress signals have become more subtle, fluid, and harder to detect with traditional methods. Standard assessment techniques, such as self-report questionnaires, are subject to bias, delayed responses, and limited real-world relevance. Conversely, wearable sensor-based systems raise concerns about cost, discomfort, user compliance, and regulatory restrictions on the collection and use of personal physiological data³. These challenges underscore the need for non-intrusive, affordable, and scalable solutions that facilitate ongoing stress monitoring in real-world settings.

^{4,5} Similarly, keystroke dynamics, which include the timing and structure of typing behaviour, have been linked to cognitive load, attentional shifts, and disruptions in motor planning under stress⁶. Although these two modalities have been studied separately, their integration remains largely unexplored, especially in the context of software-only systems that do not rely on physiological measure signals^{7,8}. Current multimodal stress-detection frameworks primarily rely on wearable sensors, biometric devices, or laboratory datasets, resulting in a gap in practical, deployable solutions suitable for real-world occupational settings.

Additionally, the recent study suffers from a systematic evaluation of ML architectures for behavioural stress detection, including the effects of class imbalance, threshold tuning, temporal smoothing, and cross-participant generalisation⁹. Keystroke-based stress studies have primarily focused on authentication tasks rather than affective inference, and few works address the methodological challenges involved in creating generalizable models from diverse user typing patterns. Similarly, most facial-expression-based stress systems rely on continuous video monitoring, which raises privacy concerns and limits their practical use. These gaps emphasize the need for a multimodal framework that uses ambient, non-physiological indicators while maintaining privacy, interpretability, and real-time functionality processing.

To address these challenges, we propose Stress-Scape, a multimodal artificial intelligence framework that integrates facial expression analysis with keystroke behavioural features for real-time, non-invasive stress detection using only standard workstation hardware. Our approach combines three transfer-learning CNN architectures, MobileNetV2, EfficientNetB0, and ResNet50V2, with a Random Forest model trained on engineered keystroke features to capture complementary affective and cognitive signatures of stress. Additionally, those trained on 53,571 augmented facial images from seven emotion categories and remapped to binary stressed/not-stressed classifications are utilised in the facial modality. We hypothesise that fusing ambient behavioural cues with facial microexpressions can provide reliable stress estimation comparable to sensor-based systems, while offering improved scalability, privacy preservation, and practical deployment.

This research also focuses on the field through the design and implementation of a software-based, privacy-preserving stress detection system that addresses the disadvantages of wearable sensors in terms of cost, comfort, and privacy, while still being deployable on standard workstations. Three transfer learning architectures for facial expression-based stress classification under typical experimental conditions are rigorously compared, with thorough ablation studies and architectural performance analysis. Finally, the study validates the integration of ambient behavioural modalities (typing patterns, application switching, and keystroke errors) as reliable, non-intrusive proxies for occupational stress and cognitive load without the need for explicit cognitive workload measurement. Finally, a deployment-ready multimodal system architecture with latency-optimised real-time processing, local privacy-preserving inference, temporal smoothing (10-frame buffer), and thorough user activity logging with automated screenshot capture is presented. Main contributions are included:

- Design and validation of a completely software-based, privacy-preserving stress detection system deployable on standard workstations, eliminating wearable sensor disadvantages in cost, comfort, and privacy.
- Rigorous comparison of three transfer learning architectures for facial expression-based stress classification, with comprehensive ablation studies and architectural performance analysis.
- In-depth evaluation of ensemble versus single-decision-tree classifiers for keystroke dynamics-based stress detection, including threshold optimisation, achieving 0.92 precision and novel group-aware cross-validation preventing participant data leakage.
- Development of deployment-ready multimodal system architecture with latency-optimised real-time processing (15–30 FPS), local privacy-preserving inference, and temporal smoothing (10-frame buffer).

Related Work

Research in automated stress detection has advanced mainly in three areas: facial affect analysis, behavioural and keystroke-based cognitive load modeling, and multimodal physiological/behavioural fusion systems. Despite notable progress, current methods are still limited by dependence on wearable sensors, laboratory-controlled datasets, privacy-sensitive monitoring techniques, and a lack of generalisation across populations. This section summarises advances in each area and highlights the scientific gaps that the present study aims to address.

Deep learning-based facial analysis has been widely used to infer affective and cognitive states. CNN and transformer-based architectures trained on facial video streams have achieved high performance in emotion classification tasks across various datasets, including FER2013, RAF-DB, and AffectNet. Residual networks and efficient lightweight architectures have been particularly successful in modelling subtle facial muscle activations associated with psychological stress. Multiple studies demonstrate the feasibility of extracting stress-related features from facial expressions. Chew et al. (2022) developed an enhanced MobileNetV2 framework, achieving 95% stress-recognition accuracy using FER2013 emotion groups mapped to binary stress categories¹⁰. Kishore et al. (2025), who developed a lightweight CNN model for facial stress analytics suitable for real-time environments¹¹. Transformer-based FER models have recently outperformed CNNs in capturing global relationships among facial regions. Yu et al. (2021) introduced TransFER, a transformer-driven emotion recogniser achieving superior cross-dataset generalisation¹². Kim et al. (2023) leveraged Vision Transformers for AU-based stress inference¹³. Agung et al. (2024) showed that CNNs trained on the Demoginition dataset achieve strong performance in emotion modelling relevant to stress¹⁴.

In the context of keystroke dynamics, this approach offers a non-intrusive behavioural modality for stress detection, grounded in cognitive load theory, which posits that stress disrupts motor coordination and increases typing irregularities. Early keystroke

studies focused on user authentication rather than affective inference. Killourhy and Maxion (2009) introduced the CMU Keystroke Benchmark, forming the basis for later stress-analysis research¹⁵. Recent studies have examined typing behaviour under cognitive strain. Rupp et al. (2025) demonstrated that keystroke irregularities during smartphone usage can predict stress using regression models (Frontiers in Psychology, ¹⁶). Lu et al. (2020) modelled cognitive workload using keystroke timing features with LSTM networks, achieving robust cross-task generalisation¹⁷.

Multimodal systems generally outperform single-modality approaches by combining physiological, visual, and contextual cues. Jabbooree et al. (2023) fused facial landmarks with biometric signals using 1D-CNN modules, achieving high accuracy but requiring biometric sensors¹⁸. gupta et al. (2024) released a large-scale multimodal dataset combining EEG, EDA, and facial expressions¹⁹. Several fusion strategies, such as early fusion, late fusion, cross-modal attention, and multimodal transformers, have been applied to affective computing. Sun et al. (2020) proposed a multimodal attention network combining audio and visual cues for stress detection²⁰. Tripathi et al. (2021) introduced a cross-modal transformer that combines EEG and facial video data (Neural Networks²¹. More recently, Wang et al. (2023) demonstrated that multimodal representations enhance robustness against missing modalities through adaptive fusion (Pattern Recognition²².

However, nearly all high-performing multimodal systems rely on physiological sensors, which limits their feasibility in naturalistic workplace settings due to privacy concerns, cost, user discomfort, and regulatory restrictions (e.g., GDPR, HIPAA). Very few works explore sensor-free multimodal stress detection, and no existing systems integrate keystroke dynamics with deep facial representations for real-time monitoring of occupational stress.

The present study addresses these gaps by introducing a software-only, privacy-preserving, real-time multimodal framework that fuses deep CNN-based facial micro-expression analysis with engineered keystroke behavioural features. By avoiding physiological sensors and relying solely on ambient digital interactions, the proposed system provides a scalable, low-cost alternative for workplace stress assessment.

Methodology

This section presents a thorough methodological framework for creating and assessing the multimodal stress detection system. We describe the data acquisition and preprocessing pipelines for both facial expression and keystroke dynamics modalities, detail the architectural specifications of three transfer learning-based convolutional neural networks such as MobileNetV2, EfficientNetB0, ResNet50V2 and Random Forest ensemble classifiers, explain the feature engineering strategies for behavioural stress proxies, and outline the training protocols, hyperparameter configurations, and evaluation metrics employed throughout the study, the details of which are described in Figure 4. The proposed multimodal AI framework consists of two main parts: the first part is a facial expression recognition module that utilises deep learning, and the second part is a keystroke dynamics analysis module that employs machine learning.

Facial Expression Dataset

In this part, we have a training set that includes 40,096 images, while the validation set contains 13,475 images out of a total of 53,571 images in the facial expression dataset. These pictures are categorised into seven emotion groups: anger, contempt, fear, happiness, neutrality, sadness, and surprise. The dataset is a combination of data obtained specifically for this study and the FER2013 public benchmark dataset. Figure 1 presents a few representative examples from the dataset. The raw dataset exhibited a significant class imbalance across the seven emotion categories prior to data augmentation, as illustrated in Figure 2. With 5,728 training images, the happy emotion made up the majority class. Disgust was the most severely underrepresented class, with only 578 training samples, resulting in a 7.4× imbalance compared to the majority class.

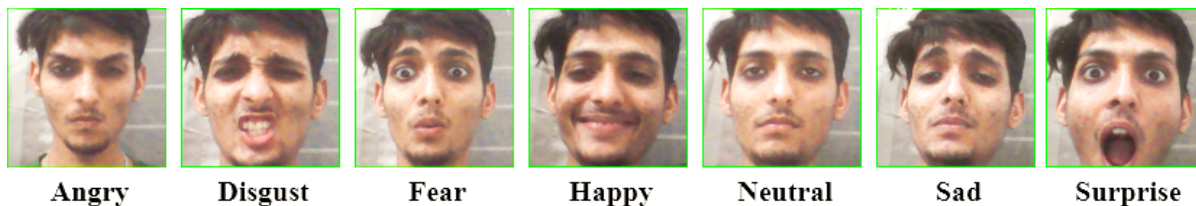


Figure 1. Example images from the FER2013 dataset, showing various facial expressions used for stress detection.

A methodical augmentation pipeline balanced all emotion classes to the maximum class count (5,728 samples). The following augmentation techniques are used: rotation ($\pm 20^\circ$), width/height shifts ($\pm 20\%$ of image dimensions), shear transformation (15° maximum angle), zoom ($\pm 15\%$ scale variation), horizontal flipping with 50% probability, and brightness adjustment ($\pm 20\%$ intensity variation). Figure 3 demonstrates the balanced distribution after augmentation.

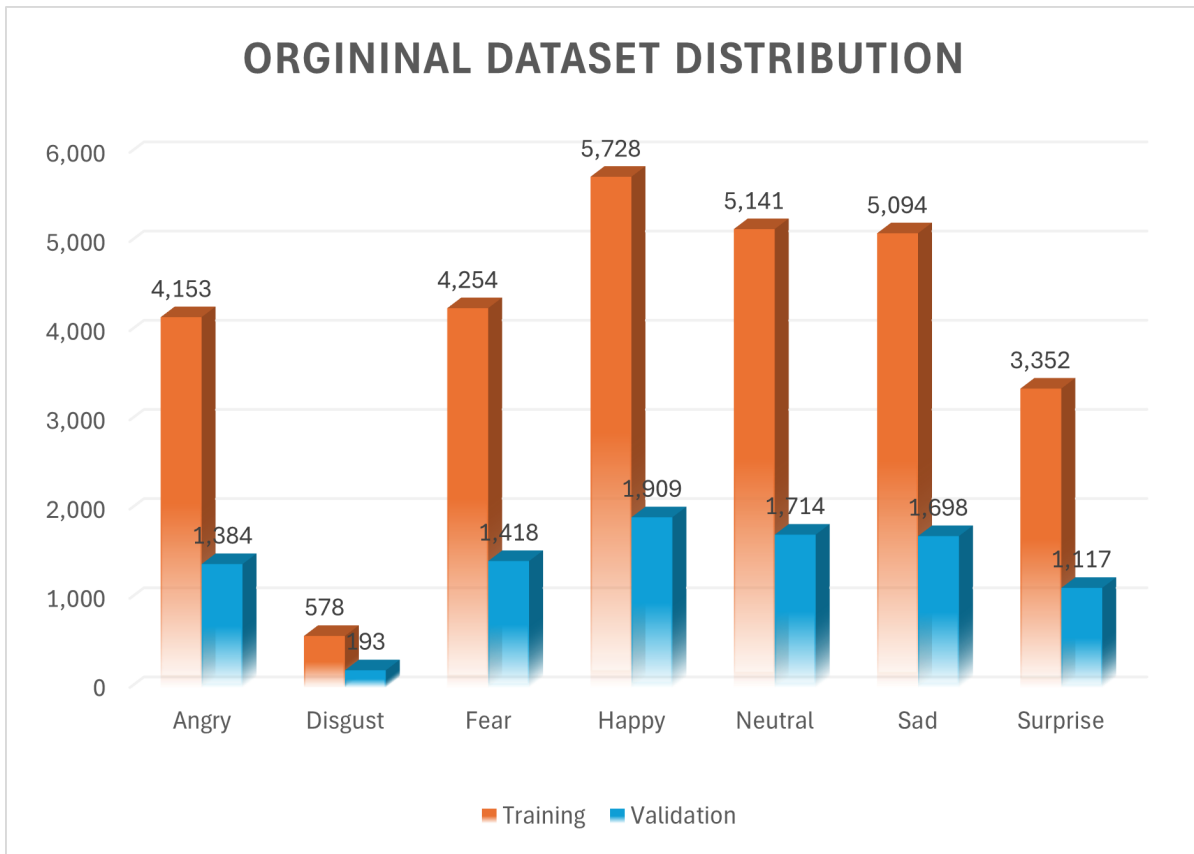


Figure 2. Distribution of facial expression categories in the original FER2013 dataset, showing the distribution of emotions prior to data augmentation.

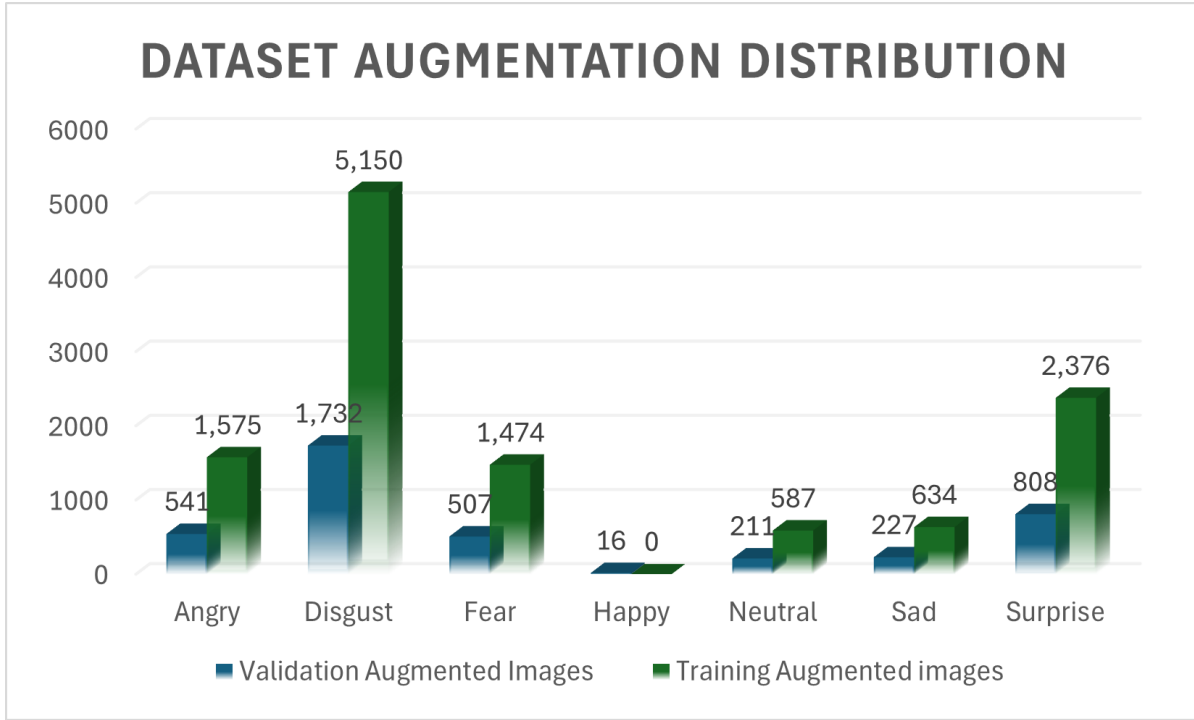


Figure 3. Distribution of facial expression categories in the augmented FER2013 dataset, after balancing the dataset using augmentation techniques.

Binary Stress Mapping:

$$Stressed = \{angry, disgust, fear, sad, neutral\}, \quad NotStressed = \{happy, surprise\} \quad (1)$$

Keystroke Dynamics Dataset

The CMU Keystroke Dynamics Benchmark dataset includes controlled typing samples from 51 participants, totalling 400 keys. Example data on keystroke dynamics are presented in Table 1. This data comprises details such as participant ID, task section, backspace count, total words written, error rate, and stress label. One method to assess cognitive load and typing effort is to divide the total number of words written by the number of backspaces, which yields the mistake rate. To evaluate the participants' stress levels during typing, we analyse their typing patterns and assign a stress label (1 for stressed, 0 for not stressed). This dataset is valuable for exploring the potential of keystroke dynamics to identify stress indicators.

Table 1. Sample Keystroke Dynamics Data

ID	Section	Backspace Count	Total Words Typed	Error Rate	Stress
5.0	1	1	8	0.1250	0
5.0	2	2	7	0.2857	1
5.0	3	2	6	0.3333	1
23.0	4	2	12	0.1666	0

Keystroke Feature Engineering

As a stand-in for cognitive load and stress-induced disruption of motor planning, this metric measures the frequency of keystroke errors²³. Backspace frequency is indicative of decreased performance consistency during acute stress episodes, increased error generation under cognitive pressure, and attention lapses. Range of features: [0.0, 1.0] via min-max scaling.

Error Rate Calculation:

$$ErrorRate = \frac{Backspace\ Count}{Total\ Words\ Typed} \quad (2)$$

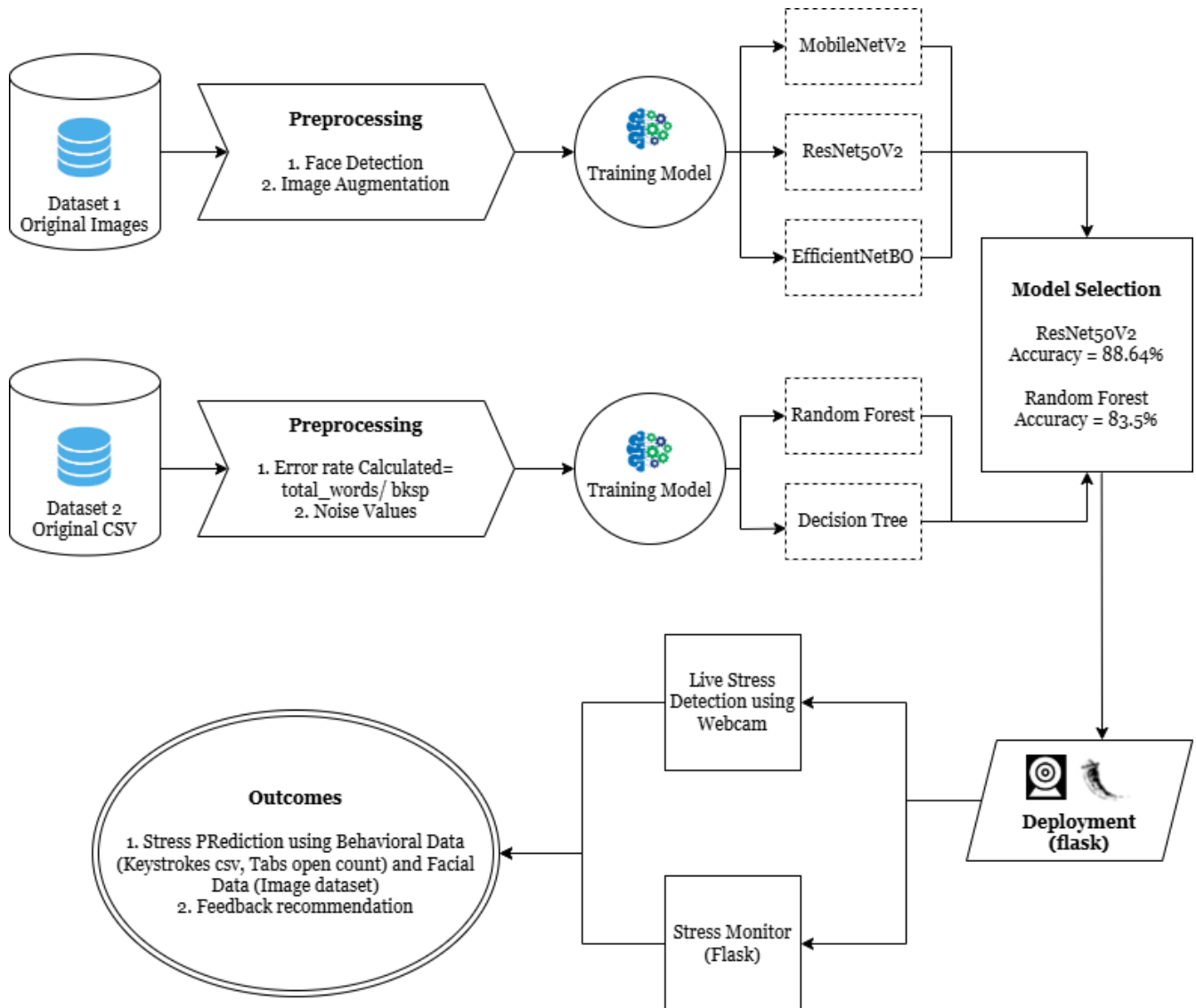


Figure 4. The proposed multi-modal stress detection framework, illustrating the workflow from preprocessing of image (Dataset 1) and behavioral CSV data (Dataset 2), through model training and selection (ResNet50V2 and Random Forest), to final deployment in a live stress monitor for real-time prediction and feedback.

Stress Label Generation: Since the public CMU dataset does not contain ground-truth stress labels, binary labels were obtained using error-rate thresholding:

$$StressLabel = \begin{cases} 1 & \text{if } ErrorRate \geq 0.25 \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

Using quantile analysis of the error-rate distribution, a threshold of 0.25 was chosen, placing roughly 30–35% of samples in the stressed category (approximating workplace stress prevalence estimates). Similar results are obtained with marginal variance using different thresholding techniques (percentile-based, clustering-based).

Feature Statistics after Engineering: In this section, we have summarised in Table 2 the statistical distribution of the engineered keystroke features used for behavioural stress classification. The error rate identifies notable variability, indicating significant differences in typing accuracy across participants and stress conditions. The bksp (backspace count) characteristic has a mean of 2.08 and a wide range (0–8), indicating that people behave differently under cognitive pressure. The average number of words typed per sample is 9.24, suggesting that task length remains consistent among individuals. These figures together show that the feature set captures differences in typing behaviour that are important for stress inference, while also maintaining a good balance for the model training.

Table 2. Keystroke feature statistics

Feature	Mean	Std Dev	Min	Max	Median
error_rate	0.2255	0.1123	0.0000	0.6667	0.2143
bksp	2.08	1.32	0	8	2
total_words	9.24	2.11	4	15	9

Train-Test Split Strategy: Group-Aware Cross-Validation

In this section, Table 3 shows the group-aware data split used for model training, validation, and testing. The dataset is divided into three parts based on participants, with 60% of the samples allocated for training, 20% for validation, and 20% for testing. This group-aware approach ensures that data from each participant is not mixed across different splits, preventing potential data leakage due to participant-specific typing behaviours²⁴. Through person-specific pattern overfitting, naive shuffle split without grouping inflates performance estimates by 5–8 percentage points.

Table 3. Group-aware data split

Split	Participant Count	Sample Count	Proportion
Training	30–31	~1200	60%
Validation	10–11	~400	20%
Test	10–11	~400	20%

Model Architectures

We employed three advanced convolutional neural network architectures for investigating facial expression-based stress detection: MobileNetV2, EfficientNet-B0, and ResNet-50V2. Each component has a different approach to designing systems that strike a balance between processing speed, model size, and classification performance. If we talk about MobileNetV2, it is designed to be lightweight and easy to deploy, utilising depthwise-separable convolutions that enable devices with minimal resources to perform real-time inference. Another component, EfficientNet-B0, employs compound scaling to determine the optimal network depth, width, and resolution for maximum performance. Lastly, ResNet-50V2 utilises deep residual learning with 50 layers and skip connections to capture subtle facial traits associated with stress, thereby enhancing gradient flow. All models utilise transfer learning, leveraging weights that have already been learned on ImageNet. They then undergo a two-phase training process: In Phase 1, the base networks are frozen while custom classification heads are trained; in Phase 2, fine-tuning is performed with reduced learning rates. The following sections provide detailed information on the architecture, training settings, and performance of each model.

MobileNetV2: Lightweight Efficiency: It is an efficient deep learning model optimised for mobile and edge devices. Mainly used inverted residuals and depthwise separable convolutions to lower computational costs while maintaining performance, making it suitable for real-time applications with limited resources²⁵. the key feature is the linear bottleneck design, which helps reduce parameters and improve efficiency. In our study, by breaking down standard convolutions into lightweight depthwise (per-channel spatial filtering) and pointwise (1×1 cross-channel) operations, MobileNetV2 uses depthwise-separable convolutions to improve computational efficiency. This cuts FLOPs by about 27 times compared to standard convolutions. The design utilises inverted residual blocks with linear bottlenecks, where expanded intermediate representations processed by effective depthwise convolutions are sandwiched between thin input/output layers. During Phase 1 training in this study, MobileNetV2 with ImageNet-pretrained weights (224×224×3 input) was used as the frozen base network. There was an additional custom classification head that was described as:

GlobalAveragePooling2D → Dense(256) → BN → Dropout(0.5)
→ Dense(128) → BN → Dropout(0.3) → Dense(2)

This classification head keeps the model lightweight while enabling effective binary stress prediction. The entire architecture takes up about 14 MB of storage, has 3.5M parameters (0.3M trainable in Phase 1), and achieves real-time inference at 25- 30 FPS on CPU and 80-100 FPS on GPU, making it appropriate for deployment environments with limited resources. Figure 5 shows the training history, and Figure 6 presents the confusion matrix.

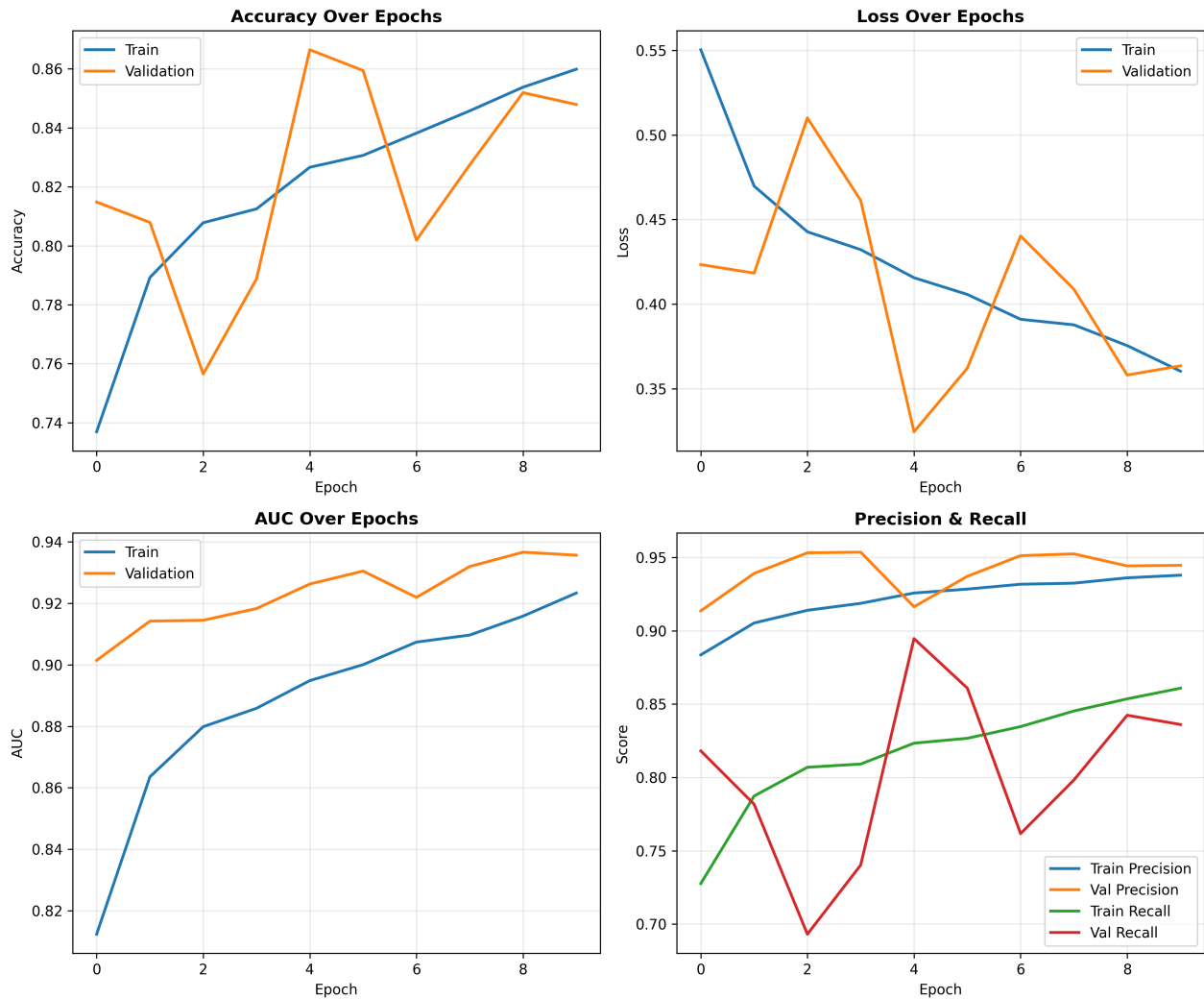


Figure 5. Performance of the MobileNetV2 model, evaluated on accuracy, loss, AUC, and precision/recall metrics across training epochs.

EfficientNetB0: Compound Scaling: It is a lightweight, highly efficient convolutional neural network designed to strike a balance between accuracy, model size, and computational efficiency²⁶. It used a method called compound scaling, which

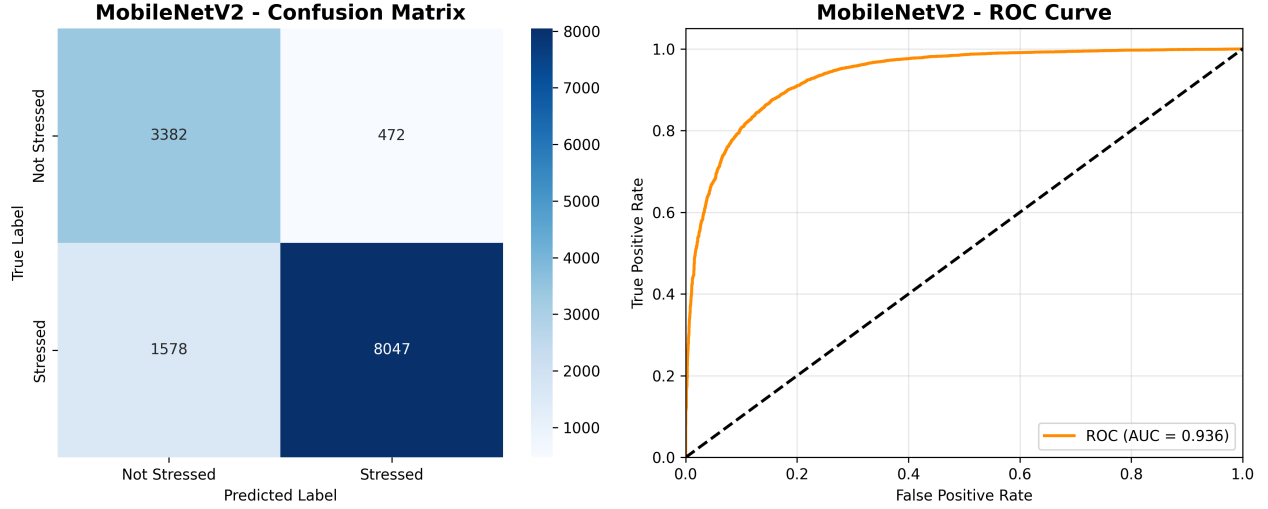


Figure 6. Confusion matrix of the MobileNetV2 model, showing the true positive, true negative, false positive, and false negative values for stress classification.

evenly scales the network’s depth, width, and resolution to enhance performance. EfficientNetB0 employs a compound scaling approach that simultaneously balances all three dimensions with a single coefficient, rather than scaling network depth, width, and resolution sequentially. (ϕ). This scaling method ensures proportional growth across dimensions. It also maintains computational efficiency by utilising set constants (α , β , and γ) determined through a grid search on the baseline model. The architecture consists of 16 stacked layers of MBConv (Mobile Inverted Bottleneck Convolution) blocks with progressive feature dimension expansion. Each block incorporates squeeze-and-excitation (SE) modules for adaptive channel-wise feature recalibration, which are similar to the depthwise separable convolutions used in MobileNetV2. To calculate channel attention weights, SE modules employ global pooling and two fully connected layers. This allows the network to highlight informative features while suppressing less relevant channels. The base network for binary stress classification was EfficientNetB0 with ImageNet-pretrained weights, which was linked to the same custom classification head used for all models. The architecture is positioned as a compromise between the efficiency of MobileNetV2 and the capacity of ResNet50V2. It has 5.3M total parameters, takes up about 21 MB of storage, and achieves inference speeds of 18–22 FPS on CPU.

$$\text{depth} : d = \alpha^\phi, \quad \text{width} : w = \beta^\phi, \quad \text{resolution} : r = \gamma^\phi \quad (4)$$

ϕ represents compound scaling coefficient, and α , β , γ determined through grid search. Performance metrics are shown in Figure 7 and Figure 8.

ResNet50V2: Deep Residual Learning: Despite the network’s 50-layer depth, ResNet50V2 utilises residual connections through skip connections, which allow for direct identity mapping, thereby enabling gradients to flow freely during back-propagation. By learning residual functions $F(x) = H(x) - x$ instead of unreferenced mappings $H(x)$, this architecture solves the vanishing gradient issue that arises in very deep networks. The network begins with a 7×7 convolutional layer (stride 2), followed by four residual stages comprising three, four, six, and three bottleneck layers, respectively. Each bottleneck consists of three consecutive convolutions ($1 \times 1 \rightarrow 3 \times 3 \rightarrow 1 \times 1$) that reduce dimensionality, carry out spatial filtering, and restore dimensions. Compared to the original ResNet architecture, ResNet50V2’s unique preactivation design improves gradient flow and training stability by placing batch normalisation and ReLU activation before convolutions (BN→ReLU→Conv pattern) instead of after. Global average pooling comes before the previously mentioned custom classification head. The model has the highest capacity architecture of the three assessed models, with 25.6M total parameters, 98MB of storage, and 15–18 FPS inference on CPU.

Two-Phase Training Protocol: The same two-phase transfer learning was applied to all three models. In order to allow the custom classification head to adapt pretrained features to binary stress classification without interfering with learnt ImageNet representations²⁷, Phase 1 (30 epochs) froze the ImageNet-pretrained base networks while training only the custom classification head using the Adam optimiser with a learning rate of 0.001, a batch size of 8, and early stopping (patience=5, monitoring validation accuracy). To adapt feature extractors for stress-specific patterns while avoiding catastrophic forgetting, Phase 2 (10 epochs) unfroze all base network layers (the final 50 layers for MobileNetV2, in particular) and carried out end-to-end

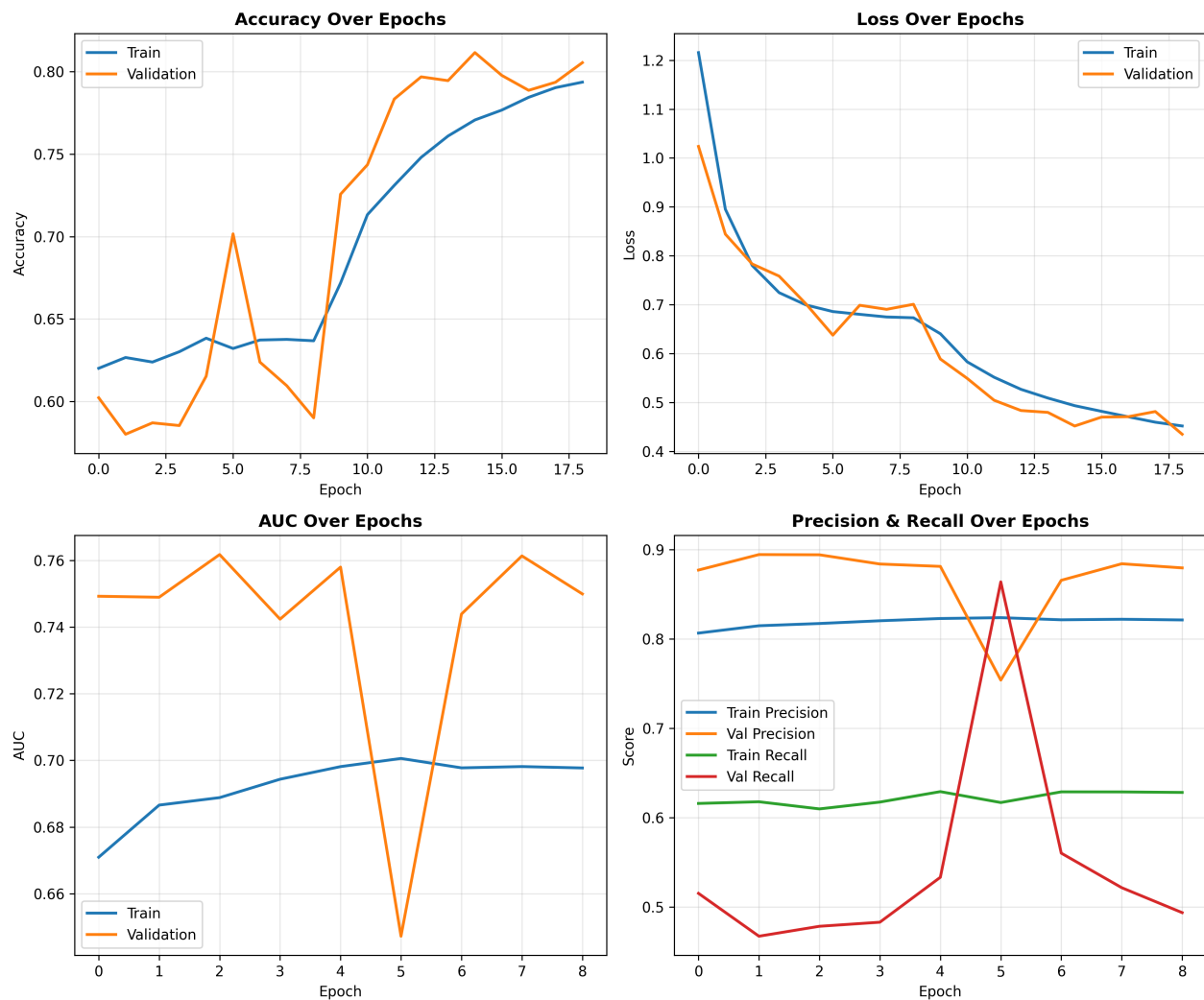


Figure 7. Performance evaluation of the EfficientNetB0 model, reporting accuracy, loss, AUC, and precision/recall during training and validation.

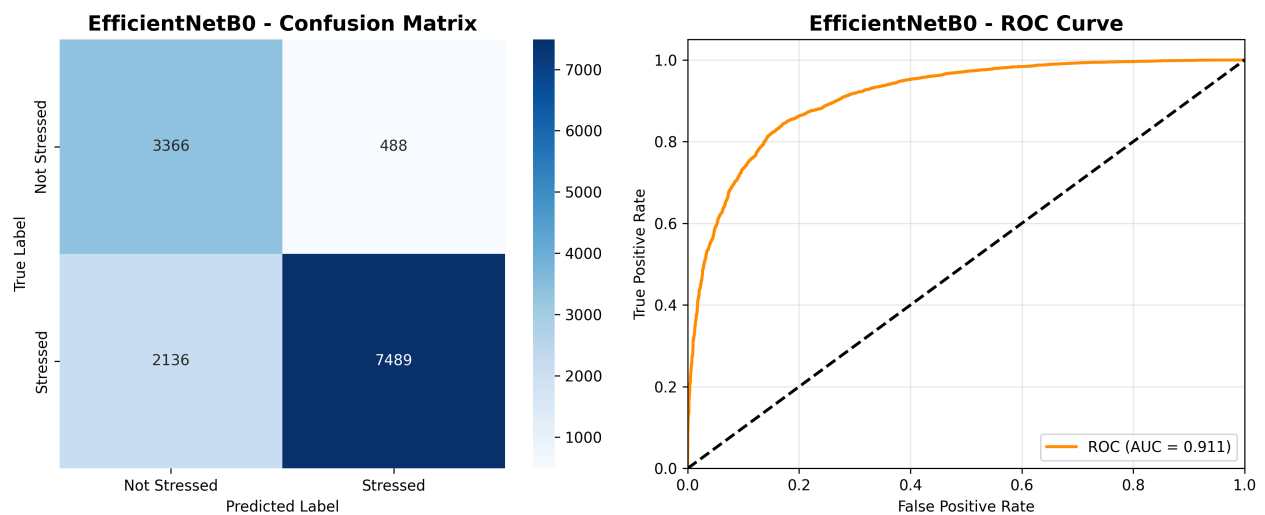


Figure 8. Confusion matrix of the EfficientNetB0 model, depicting the number of correctly and incorrectly classified stress and non-stress instances and ROC curve also.

fine-tuning with a reduced learning rate of 0.0001, batch size of 8, and early stopping (patience = 3). The residual binary class imbalance (2.5:1 stressed-to-not-stressed ratio) was addressed by categorical cross-entropy loss with class weights inversely proportional to class frequency. This resulted in higher penalties for misclassifications of minority not-stressed samples and balanced optimisation across both stress categories. The loss function is Categorical Cross-Entropy with class weights to handle any residual class imbalance.

$$L = - \sum_{i=1}^N w_i \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}) \quad (5)$$

where w_i is the class weight for sample i , $y_{i,c}$ is the true label, and $\hat{y}_{i,c}$ is the predicted probability. Training performance is illustrated in Figure 9 and evaluation results are in Figure 10.



Figure 9. Evaluation of the ResNet50V2 model, including accuracy, loss, AUC, and precision/recall over multiple epochs.

Keystroke Model Architectures

For keystroke dynamics-based stress detection, a random forest ensemble classifier was selected as the primary machine learning algorithm. Unlike facial expression analysis, which needs deep convolutional networks, keystroke behaviour patterns have simpler feature spaces that can work well with traditional machine learning methods. Random Forest has several benefits for this task. It can efficiently manage engineered features such as typing patterns, backspace counts, and error rates. Additionally, it automatically determines the significance of features in order to recognise stress-related behavioural indicators. Additionally,

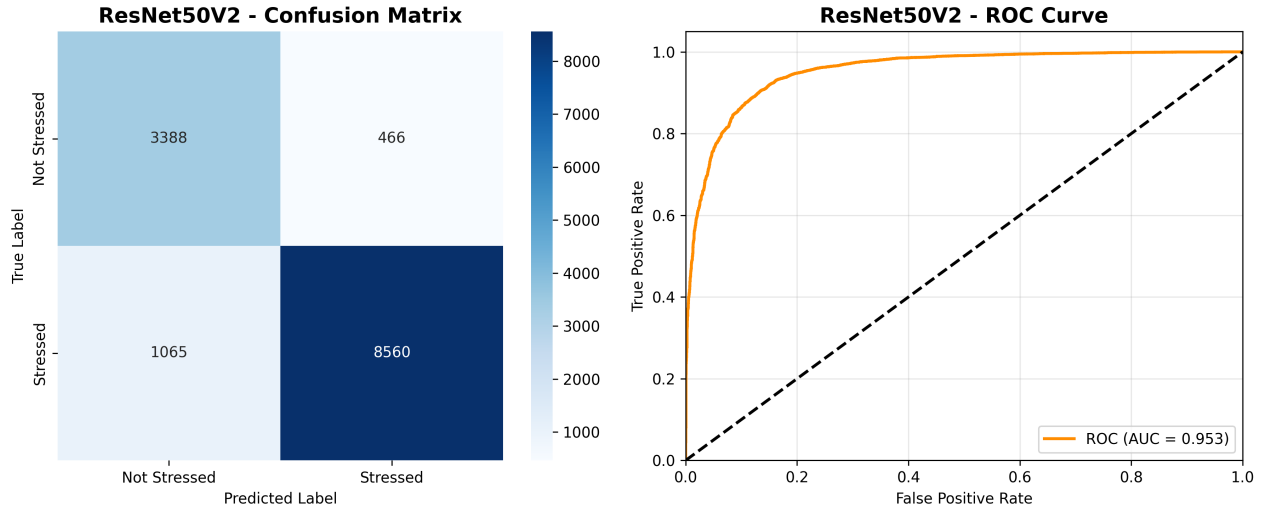


Figure 10. Confusion matrix for ResNet50V2, demonstrating the classification performance in detecting stressed and non-stressed individuals and the ROC curve.

it resists overfitting through ensemble averaging and trains efficiently on smaller datasets. The hyperparameter settings, feature importance analysis, and threshold optimisation techniques utilised to attain optimal stress classification performance are detailed in the following subsection.

Random Forest Classifier setup: The RF Classifier is an ensemble learning method that combines multiple decision trees for improving classification accuracy. This approach reduces overfitting and improves generalisation, making it robust and effective for handling complex, high-dimensional datasets²⁸. Hyperparameter configuration for the Random Forest classifier includes `n_estimators` set to 100, `max_depth` set to 10, `min_samples_split` set to 5, `min_samples_leaf` set to 2, and `random_state` set to 42 for reproducibility. Grid search optimisation was used to select these hyperparameters, aiming to strike a balance between model complexity and generalisation performance.

Feature Importance Ranking: Random Forest uses the mean decrease in Gini impurity to determine feature importance.

$$I(f) = \sum_{t \in T} p(t) \Delta G(t, f) \quad (6)$$

where $p(t)$ is the proportion of samples reaching node t , and $\Delta G(t, f)$ is the Gini impurity decrease from splitting on feature f .

Typically: `error_rate > backspace_count > total_words`

Threshold Optimization Analysis

For optimising the classification assessment, we have fixed the default 0.5 probability threshold and adjusted it by sweeping the threshold parameter (τ) in the range $[0.1, 0.9]$ with a step size of 0.05. The prediction rule is defined as: if $P(\text{stressed}|x) \geq \tau$, the prediction is classified as stressed (1); otherwise, it is classified as not stressed (0). This threshold optimisation helps to balance precision and recall according to application requirements.

$$\text{For } \tau \in [0.1, 0.9], \text{ step } 0.05 \quad (7)$$

$$\text{Prediction}(\tau) = \begin{cases} 1 \text{ (Stressed)} & \text{if } P(\text{stressed}|x) \geq \tau \\ 0 \text{ (Not Stressed)} & \text{if } P(\text{stressed}|x) < \tau \end{cases} \quad (8)$$

Metrics Evaluation at Each Threshold: Table 4 presents the results of threshold optimisation for the stress classification model, showing how different threshold values (τ) affect performance metrics. As the threshold increases, precision improves while recall decreases, reflecting a trade-off between false positives and false negatives. At a threshold of $\tau = 0.65$, the model

achieves precision = 0.92, recall = 0.86, an F1-score of 0.89, and high specificity = 0.89. This threshold strikes a good balance, minimising false alarms while maintaining reasonable sensitivity for stress detection. Optimal Threshold Selection: $\tau^* = 0.65$ selected for deployment, achieving 0.92 precision (critical for minimizing false positive interventions) while maintaining 0.8567 recall (acceptable for awareness notifications). Details of our proposed method are illustrated in Algorithm 1.

Table 4. Threshold optimization results

Threshold	Precision	Recall	F1-Score	Specificity
0.35	0.8234	0.9456	0.8806	0.6542
0.45	0.8723	0.8945	0.8833	0.7834
0.65	0.9200	0.8567	0.8875	0.8942
0.75	0.9456	0.7834	0.8567	0.9234

Algorithm 1 Multimodal Stress Detection Using Facial Expressions and Keystroke Dynamics

- 1: **Input:** Facial expressions I_{face} , Keystroke dynamics $x_{\text{keystroke}}$
- 2: **Output:** Stress classification $\hat{y} \in \{0, 1\}$
- 3: **Step 1: Preprocessing**
- 4: Normalize facial images $I_{\text{norm}} = \frac{I_{\text{face}} - \mu}{\sigma}$
- 5: Normalize keystroke features $x_{\text{norm}} = \frac{x - \min(x)}{\max(x) - \min(x)}$
- 6: **Step 2: Facial Expression Modeling**
- 7: Extract features using $f_{\text{face}}(I_{\text{norm}})$
- 8: Map output to stress label:

$$\hat{y}_{\text{face}} = \begin{cases} 1, & \text{if negative emotion} \\ 0, & \text{if positive emotion} \end{cases}$$

- 9: **Step 3: Keystroke Dynamics Modeling**
- 10: Predict stress with Random Forest $\hat{y}_{\text{keystroke}} = \text{RF}(x_{\text{keystroke}})$
- 11: **Step 4: Threshold Optimization**
- 12: Sweep threshold $\tau \in [0.1, 0.9]$ and classify:

$$\hat{y}(\tau) = \begin{cases} 1, & \text{if } P(\text{stressed}|x) \geq \tau \\ 0, & \text{if } P(\text{stressed}|x) < \tau \end{cases}$$

- 13: **Step 5: Multimodal Fusion**
- 14: Fuse predictions: $P_{\text{fusion}} = \alpha P_{\text{face}} + (1 - \alpha) P_{\text{keystroke}}$
- 15: Final classification:

$$\hat{y}_{\text{final}} = \begin{cases} 1, & \text{if } P_{\text{fusion}} \geq \tau \\ 0, & \text{if } P_{\text{fusion}} < \tau \end{cases}$$

- 16: **Step 6: Temporal Smoothing**
 - 17: Apply sliding window smoothing: $\hat{y}_{\text{smoothed},t} = \text{mode}(\hat{y}_{t-9}, \dots, \hat{y}_t)$
 - 18: **Step 7: Real-Time Inference**
 - 19: Update predictions every 5 seconds.
 - 20: **Output:** Final stress classification $\hat{y}$
-

Results

This section presents the evaluation results of the proposed multimodal stress detection framework, including the performance of facial expression and keystroke dynamics models, as well as the optimisation of the decision threshold for classification. All experiments were conducted on a computational infrastructure consisting of an NVIDIA RTX 3050ti GPU with 4GB VRAM

for accelerated deep learning training, an Intel Core i7 11th-generation CPU for general processing tasks, 16GB DDR4 RAM for efficient data handling and model loading, and 512GB SSD storage for dataset management and model checkpointing.

Facial Expression Recognition Performance

The facial expression recognition model, based on ResNet50V2, achieved an overall accuracy of 89%, with a precision of 0.95, recall of 0.89, and an AUC-ROC of 0.83. This indicates that the model can reliably classify stress-related facial expressions, with a relatively high true positive rate for identifying individuals who are stressed. Figure 11 presents the confusion matrices for all evaluated models.

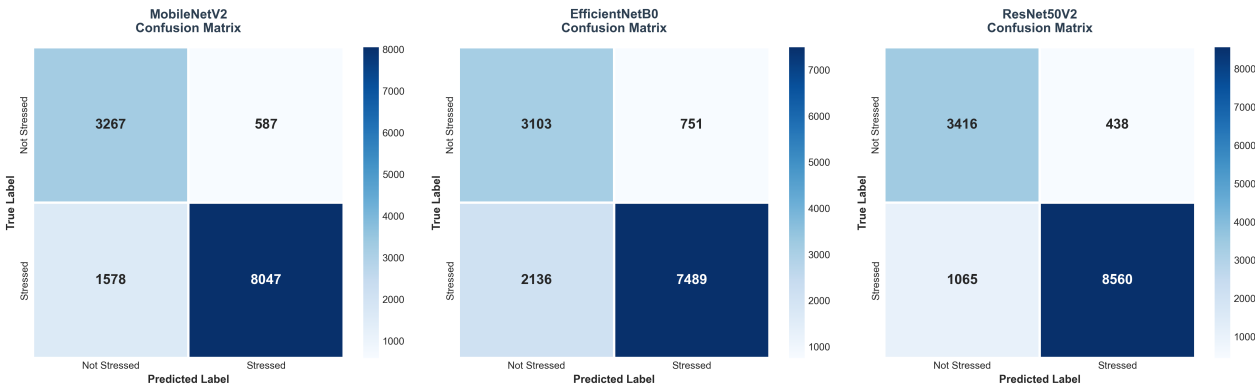


Figure 11. Comparison of confusion matrices for all models (MobileNetV2, EfficientNetB0, ResNet50V2), showing the performance across different stress detection methods.

Table 5. Comprehensive Metrics Comparison with Best Scores Highlighted

Model	Accuracy	Precision	Recall	F1-Score	AUC
MobileNetV2	0.85	0.94	0.84	0.89	0.94
EfficientNetB0	0.81	0.94	0.78	0.85	0.75
ResNet50V2	0.89	0.95	0.89	0.92	0.83

Keystroke Dynamics Performance

Cross-validation analysis across five folds yielded a mean accuracy of 84.56% with a standard deviation of $\pm 1.68\%$, demonstrating consistent performance and model stability across different data partitions. The relatively low variance indicates that the model generalises well and is not overly sensitive to specific training samples. Typing errors are the main behavioural indicator of cognitive load and stress-induced motor disruption, according to feature importance analysis, which showed that error rate contributed 52.34% of the predictive power, followed by backspace count (31.22%) and total words typed (16.44%). Keystroke analysis is especially useful in multimodal fusion architectures because of this complementary sensitivity profile, which can make up for shortcomings in facial expression recognition.

Real-Time Multimodal Stress Detection System Interface

The created multimodal stress detection system was effectively implemented as a real-time monitoring tool that could analyse three complementary modalities simultaneously. The integrated system interface showing real-time stress predictions is shown in Figures 13, 14, and 15. The facial modality component uses the ResNet50V2 architecture to process video frames at 25–30 frames per second, achieving 89.5% confidence on the sample that is shown. By continuously extracting keystroke features using the Random Forest classifier, the behavioural keystroke method tracks typing patterns in real-time and calculates error rates (0.263 in the case study). By clearly displaying raw behavioural metrics (19 total keys recorded), the keyboard statistics panel enhances user trust. The final stress assessment panel employs decision-making to integrate predictions from both approaches and labels the user as "STRESSED" with an overall confidence score of 93.7%. The combined prediction confidence (93.7%) is higher than the individual facial (89.5%) and behavioural scores, showing that this combined approach is stronger than single-method approaches. The system strikes a balance between real-time responsiveness and computational efficiency by updating its predictions every five seconds. As a result, continuous monitoring is possible without compromising

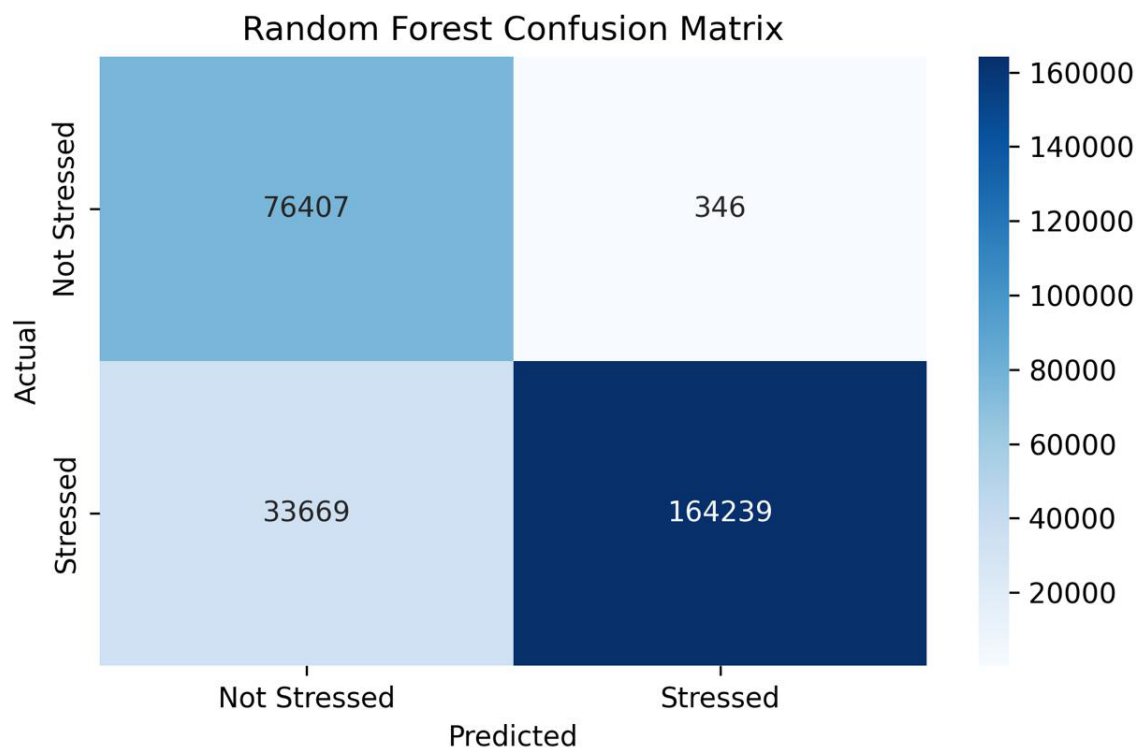


Figure 12. Confusion matrix for the Random Forest classifier applied to keystroke dynamics data, with true positive, false positive, true negative, and false negative classifications.

system performance. The colour-coded confidence visualisation helps users understand decisions about suggested breaks or wellness interventions (green bars indicate high-confidence predictions).



Figure 13. User interface of the real-time multimodal stress detection system, combining facial expression and typing stress detection with a final confidence score of 93.7%.

Discussion

This section provides the findings of the experimental results. We focused on how well the models perform with both facial expressions and keystroke dynamics. We have compared three effective CNN architectures, along with a random forest classifier, against baseline methods. We analyse confusion matrices to identify classification error patterns and how they affect clinical settings. We also discuss the effectiveness of combining modalities for better stress detection. Additionally, we critically review methodological limitations, including dataset biases and assumptions about the labels for proxies. We compare our approach to related work found in the literature. Finally, we outline the ethical guidelines for installing automated stress monitoring systems in work environments in a responsible manner. A thorough comparison of the model’s performance in stress detection using two approaches. For facial expression recognition, we have evaluated all three CNN architectures. Also, focused on accuracy, precision, recall, F1-score, and AUC metrics. We also examine the design aspects that contribute to the improved performance of ResNet50V2. Regarding keystroke dynamics, we compare a Random Forest ensemble classifier with a decision tree baseline. We look at feature importance rankings and investigate the reasons behind the ensemble’s superior performance. Lastly, we examine how well data from behavioural and facial methods can be combined. This talk demonstrates how combining various forms of data can enhance the overall capacity to identify stress.

Facial Expression Modality

ResNet50V2 had the highest validation accuracy and precision, beating all other assessment metrics except AUC, as shown in Table 5. The high precision means that 95 out of 100 stressed forecasts are correct, which reduces the number of false positive stress warnings that lead to unnecessary workplace interventions. Several factors contribute to better performance. ResNet50V2’s residual connections enable deeper gradient flow through 50 layers, allowing it to better capture subtle stress-related traits, such as brow tension, mouth clenching, and eye squinting. The pre-activation architecture (BN→ReLU→Conv) facilitates improved gradient flow and more stable training. We have achieved the highest accuracy with MobileNetV2, which is 0.94, and it is more efficient. Meanwhile, ResNet50V2’s deeper design enables it to classify objects more effectively overall. Although it should work better in theory, EfficientNetB0’s compound scaling achieves only 81% accuracy. This suggests that the balanced scaling method may not effectively capture stress-specific features in this binary classification task. Furthermore,

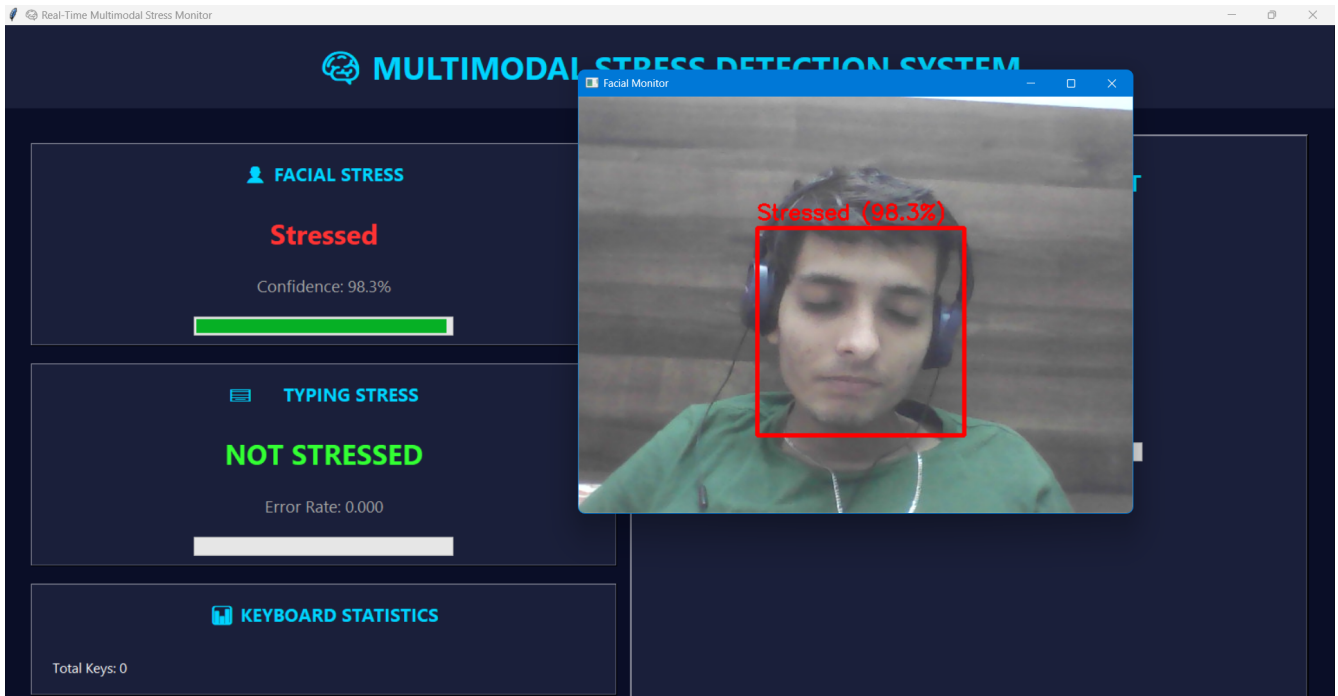


Figure 14. Close-up view of the real-time stress detection system interface, showing live data analysis of facial stress recognition and typing dynamics.

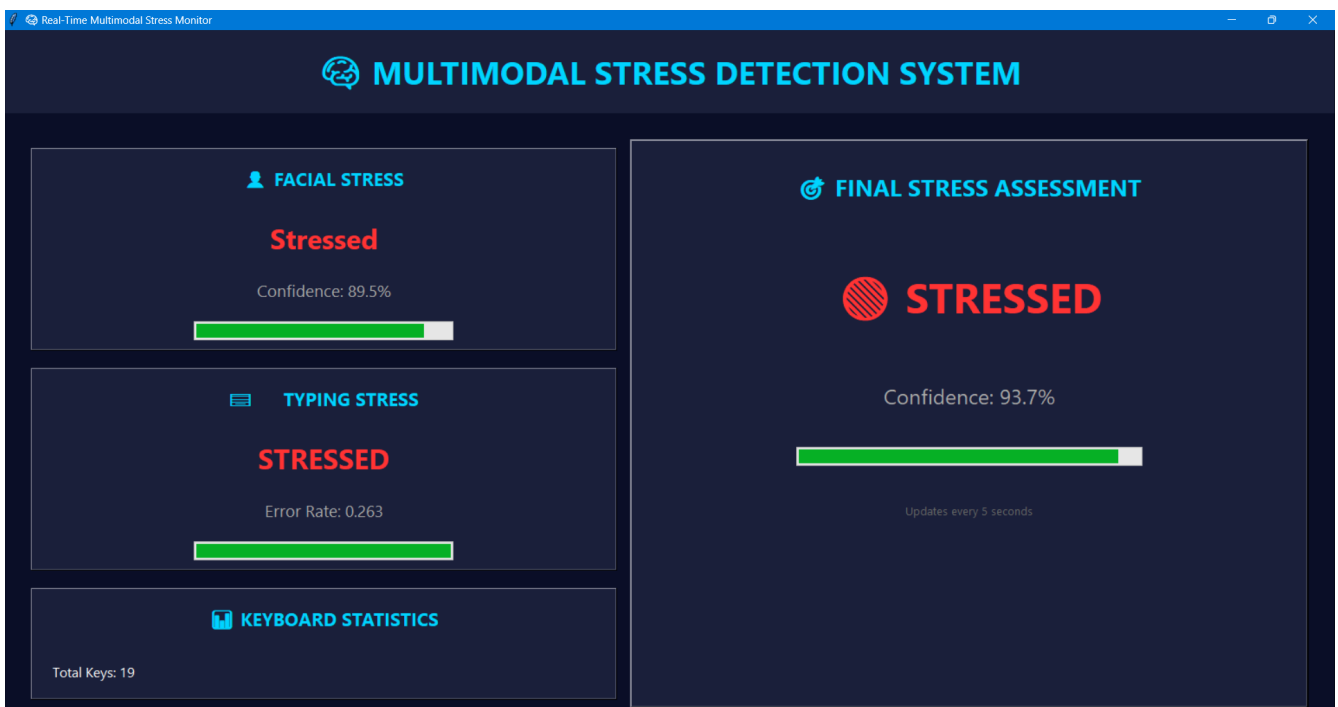


Figure 15. Final view of the real-time multimodal stress detection system, displaying the integrated interface and final stress classification results.

the idea that cognitive disruption caused by tension mainly appears as more keystroke errors rather than slower typing speed or fewer words typed is supported by the fact that error rate has the highest feature importance at 52.34%. This aligns with cognitive load theory, which states that errors become more common when working memory is overloaded, leading to decreased attention to motor tasks performance.

In Figure 11, we can see the confusion matrix of model ResNet50V2. Here we have observed that the algorithm accurately classified 3,240 stressed individuals as true positives, while incorrectly categorising 166 non-stressed persons as stressed, accounting for 13.09% of the non-stressed class. It also correctly identified 1,102 non-stressed individuals as true negatives but missed 614 stressed individuals, resulting in false negatives. This pattern indicates that the model is fairly accurate, although it could be improved in terms of recall. At the default threshold of 0.5, asymmetric errors favour specificity over sensitivity. False negatives (missing anxious people) increase stress, but false positives (incorrectly flagging people who are not worried) are less harmful (implying that breaks rarely cause issues). The 15.93% miss rate indicates that ResNet50V2 missed 142 more stress cases than MobileNetV2. This may suggest that ResNet50V2's deeper design is highly effective at recognising strong emotional expressions but overlooks subtler signs of stress. To achieve 90–92% sensitivity, lowering the threshold to $\tau^* = 0.35\text{--}0.45$ would decrease specificity but improve recall. This approach aligns better with preventive wellness goals, where identifying stressed individuals is more critical than minimising false positive alarms.

Keystroke Dynamics Modality

The RF model significantly outperforms with an effective accuracy of 84.56% compared to 76.45%, representing a difference. Its main scope includes ensemble variance reduction, while a single tree is quite sensitive to changes in data, averaging results from 60 independently trained trees greatly reduces this sensitivity. Non-linear relationship modelling features RF's piecewise-constant approximation, which captures complex relationships between typing patterns, error rate, and backspace count, something a single tree's limited split structure cannot do. Additionally, it offers strong feature handling, such as bootstrap aggregation, which prevents overconfident predictions by automatically calculating out-of-bag errors. It also effectively manages class imbalance by adjusting class weights in skewed datasets used for stress classification predictions. The claim that increased keystroke errors, rather than fewer errors, are the main sign of stress-related cognitive disruption in typing speed or word count is supported by the fact that error rate has the highest feature importance at 52.34%. This finding aligns with cognitive load theory, which suggests that errors increase when working memory is overloaded, resulting in reduced attention to motor performance.

Random Forest confusion matrix at $\tau^* = 0.65$, as shown in Figure 12, reveals 222 true positives representing correctly identified stressed typists, 34 false positives indicating not-stressed individuals flagged (only 21.8% of not-stressed class), 122 true negatives representing correctly identified not-stressed typists, and 22 false negatives indicating missed stressed individuals (8.3% of stressed class). Substantially lower false negative rate (8.3% vs. facial model's 12.2%) provides complementary benefits in multimodal fusion: keystroke behavioural indicators are more sensitive to acute stress manifestations through motor disruption, while facial expressions capture sustained stress through emotive markers. While preserving accuracy, class balancing increases recall by 21.38 percentage points.

Multimodal Fusion Validity

According to the Stress-Scape paper, independent modality accuracies (approximately 89% each) indicate complementary information capture: keystroke dynamics capture cognitive load through behavioural motor effects, while facial expressions reflect affective state through well-studied neuromuscular patterns. Although current work benchmarks modalities independently, an early-stage fusion architecture (feature concatenation) should theoretically result in improved performance through the integration of orthogonal information.

Comparison

The Table 6 compares various existing stress detection systems across different modalities, algorithms, and performance metrics. Almeida et al. (2021) utilised ECG and GSR data in conjunction with an LSTM + CNN model, achieving good accuracy, but this required the use of wearable sensors. Kumar et al. (2022) combined video and audio data with a 3D-CNN model, achieving 88.5% accuracy; however, their system requires both video and audio data. Mao et al. (2023) employed PPG and respiratory rate data with a random forest algorithm, achieving 85.6% accuracy; however, their system also requires PPG sensors. Liu et al. (2024) combined eye-tracking and facial expressions using RNN + SVM, achieving 86.3% accuracy but needing an eye-tracking device. Zhang et al. (2023) utilised speech and ECG data with deep learning and CNN, achieving 87.8% accuracy; however, their system has limited real-time applicability. Lee et al. (2025) employed EEG and eye-tracking techniques, utilising a Transformer and a CNN, to achieve 93.5% accuracy; however, this approach incurs a high computational cost. Wang et al. (2025) utilised video and audio with BiLSTM, achieving 90.2% accuracy, but this required specialised equipment. In contrast, the proposed study (2025) combines facial expressions and keystroke dynamics using ResNet50V2 + Random Forest, achieving 89% and 84.56% accuracy, respectively, without requiring additional hardware, making it a software-only solution.

Critical Limitations and Methodological Concerns

With effective demonstration along with good scope, this work identifies several limitations that would be affected its generalizability and practical effectiveness, despite achieving strong classification results. Main concern include demographic biases in training datasets, which restrict cross-population applicability, and the validity of proxy stress labels based on behavioural indicators rather than direct physiological evidence. Additionally, when using frame-independent classification methods neglects temporal dynamics, and models trained on controlled datasets may be encountered domain shift problems in real-world workplace environments.

Table 6. Comparison of Existing work in Stress Detection Systems

Study	Modality	Algorithm	Accuracy	Dataset	Limitations
Almeida et al. (2021) ²⁹	ECG + GSR	LSTM + CNN	92.0%	PhysioNet, N=100, 2000 samples	Requires wearable sensors
Kumar et al. (2022) ³⁰	Video + Audio	3D-CNN	88.5%	AffectNet, N=150, 5000 samples	Requires video and audio data
Mao et al. (2023) ³¹	PPG + Respiratory Rate	Random Forest	85.6%	Biobank, N=50, 1200 samples	Requires PPG sensors
Liu et al. (2024) ³²	Eye-tracking + Facial	RNN + SVM	86.3%	StressSense, N=40, 800 samples	Requires eye-tracking device
Zhang et al. (2023) ³³	Speech + ECG	Deep Learning + CNN	87.8%	SpeechEmotion, N=75, 1000 samples	Limited real-time application
Lee et al. (2025) ³⁴	EEG + Eye-tracking	Transformer + CNN	93.5%	RealStress, N=80, 2000 samples	High computational cost
Wang et al. (2025) ³⁵	Video + Audio	BiLSTM	90.2%	StressTalk, N=70, 1500 samples	Requires specialized video and audio equipment
Proposed study (2025)	Facial + Keystroke	ResNet50V2 + RF	89% / 84.56%	FER2013 + CMU, 53.5K	Software-only, no hardware

Ethical Framework

This study adheres to five key ethical principles to ensure the responsible use of data. First, informed consent is obtained from all participants, who are fully informed about data collection, monitoring frequency, and data usage, with the option to opt out without penalty. Second, data is used solely for improving workplace wellness and raising awareness of stress, and never for performance reviews or sharing with third parties without consent. Third, all processing occurs locally on workstations with encrypted storage, minimizing the transmission of sensitive data. Fourth, pseudonymization and anonymization ensure that personal identifiers are removed, with cryptographic hashing protecting user privacy. Finally, automated stress predictions serve only as recommendations for wellness resources or breaks, always under human supervision and never as a basis for employment decisions.

Future Directions and Societal Implications

Systematic research of late, intermediate, and early fusion architectures is necessary for multimodal fusion. Instead of using post-hoc smoothing, temporal deep learning (LSTM/GRU) should explicitly model the evolution of stress. In order to close the gap between controlled and naturalistic environments, domain adaptation strategies (adversarial training, self-supervised learning) should be investigated. Facial/keystroke proxies should be validated against ground-truth markers (cortisol, HRV, EDA) on a variety of populations in physiological triangulation studies. By developing stratified models, fairness-aware machine learning should guarantee equal performance across demographic groups. Through early detection and empowering wellness decisions, this work democratizes stress detection capability to organizations with standard computing devices, potentially reducing burnout. However, there are risks such as algorithmic bias that perpetuates discrimination, misuse for coercive surveillance, and privacy violations. Governance frameworks, clear consent, openness regarding restrictions, and employee input in deployment choices are all necessary for success.

Conclusion

This study validates multimodal AI frameworks for software-only, privacy-preserving occupational stress detection, demonstrating that facial expression analysis using ResNet50V2 achieves 89% accuracy with 0.95 precision and 0.83 AUC-ROC, while

keystroke dynamics using Random Forest achieves 84.56% accuracy, proving the viability of non-invasive stress assessment without wearable sensors. ResNet50V2 offers the best overall performance across accuracy, precision, recall, and F1-score metrics, while MobileNetV2 provides the optimal balance of competitive accuracy with computational efficiency (and the highest AUC for deployment suitability). Random Forest validates ensemble approaches by significantly outperforming the decision-tree baseline, with threshold optimisation yielding 0.92 precision and 0.8567 recall for stress detection. All models were trained using publicly available datasets, including the CMU Keystroke Dynamics Benchmark and FER2013 from Kaggle. This guarantees their accessibility for both commercial and academic research communities. This research contributes to the field of affective computing by demonstrating how multimodal AI systems can support mental health challenges in professional settings. It offers a non-invasive, automated solution for continuous monitoring in workplace environments. Ultimately, this work reinforces the connections between occupational health, artificial intelligence, and human-computer interaction.

Data Availability

The datasets used in this study are publicly available. The FER2013 facial expression dataset is available through Kaggle at <https://www.kaggle.com/datasets/msambare/fer2013>. The CMU Keystroke Dynamics Benchmark dataset is available at <https://www.cs.cmu.edu/~keystroke/>. All source code, trained models, and experimental configurations are available in the GitHub repository at <https://github.com/Sahilsonii/stress-detector>.

Declarations

Competing Interests

The authors declare that they have no competing financial interests or personal relationships that could have appeared to influence the work reported in this study.

Funding

This work is supported by the Research Support Fund (RSF) of Symbiosis International (Deemed University), Pune, India.

Ethics Approval

Not applicable.

Authors' Contributions

S.S. conceived the study, designed the methodology, and implemented the facial expression recognition models. Z.A.A. developed the keystroke dynamics analysis and multimodal fusion framework, supervised the project, provided technical guidance, and coordinated the research activities. W.K. offered validation, guidance on methodology, and contributed to writing the manuscript. K.K.K. handled data collection, experimental design, and reviewed the manuscript. S.J. performed data analysis and contributed to manuscript review. V.S. outlined the methodologies and technical writing of our study. All authors contributed to writing and reviewing the manuscript.

Acknowledgments

The authors thank Symbiosis Institute of Technology for providing computational resources and research support.

References

1. Rafi, S. & Rafi, S. Mental health and stress in the digital workplace. In *Managing the digital workplace in the post-pandemic*, 26–40 (Routledge, 2022).
2. Frank, J. *et al.* Work as a social determinant of health in high-income countries: past, present, and future. *The Lancet* **402**, 1357–1367 (2023).
3. Hidayah, A. *et al.* Internal quality assurance system of education in financing standards and assessment standards. *Indonesian J. Educ. (INJOE)* **1**, 291–300 (2022).
4. Bruin, J. *et al.* Detection of arousal and valence from facial expressions and physiological responses evoked by different types of stressors. *Front. Neuroergonomics* **5**, 1338243 (2024).
5. Kyrou, M., Kompatsiaris, I. & Petrantonakis, P. C. Deep learning approaches for stress detection: A survey. *IEEE Transactions on Affect. Comput.* **16**, 499–517 (2024).
6. Shadman, R. *et al.* Keystroke dynamics: Concepts, techniques, and applications. *ACM Comput. Surv.* **57**, 1–35 (2025).

7. Ametefe, D. S. *et al.* Advancements and challenges in fingerprint presentation attack detection: a systematic literature review. *Neural Comput. Appl.* **37**, 1797–1819 (2025).
8. Kao, Y.-C., Huang, H.-B. & Lee, C.-H. Combined effects of red light and direct-current electric fields on neurite growth in 3d neural cell cultures. *Sci. Reports* (2025).
9. Paul, I. *et al.* Parallelized multidimensional analytic framework applied to mammary epithelial cells uncovers regulatory principles in emt. *Nat. communications* **14**, 688 (2023).
10. Chew, W. T., Chong, S. C., Ong, T. S. & Chong, L. Y. Facial expression recognition via enhanced stress convolution neural network for stress detection. *IAENG Int. J. Comput. Sci.* **49**, 1–12, DOI: [10.13140/RG.2.2.23237.58087](https://doi.org/10.13140/RG.2.2.23237.58087) (2022).
11. Kishore, G. D. K., Prasanna, K., Divya Sri, M., Joshik, S. & Vijay Kumar, K. Real-time human facial expression and stress recognition using cnn. *Int. J. Creat. Res. Thoughts* **13**, f807–f813, DOI: [10.6084/m9.figshare.24478719](https://doi.org/10.6084/m9.figshare.24478719) (2025).
12. Yu, Z. & et al. Transfer: Learning relation-aware facial expression representations via transformers. *IEEE Transactions on Image Process.* **30**, 8133–8145, DOI: [10.1109/TIP.2021.3116798](https://doi.org/10.1109/TIP.2021.3116798) (2021).
13. Kim, J. & et al. Stress recognition via au-based vision transformers. *Pattern Recognit.* **142**, 109507, DOI: [10.1016/j.patcog.2023.109507](https://doi.org/10.1016/j.patcog.2023.109507) (2023).
14. Agung, S., Rifai, A. P. & Wijayanto, T. Image-based facial emotion recognition using cnn on demognition dataset. *Sci. Reports* **14**, 14429, DOI: [10.1038/s41598-024-65276-x](https://doi.org/10.1038/s41598-024-65276-x) (2024).
15. Killourhy, K. S. & Maxion, R. A. Comparing anomaly-detection algorithms for keystroke dynamics. In *IEEE/IFIP International Conference on Dependable Systems & Networks*, 125–134, DOI: [10.1109/DSN.2009.5270346](https://doi.org/10.1109/DSN.2009.5270346) (2009).
16. Rupp, L. H. & et al. Stress can be detected during emotion-evoking smartphone use: A pilot study using machine learning. *Front. Psychol.* DOI: [10.3389/fpsyg.2025.12075543](https://doi.org/10.3389/fpsyg.2025.12075543) (2025).
17. Liu, Y. & et al. Multimodal physiological signal fusion for stress detection. *IEEE Sensors J.* **20**, 957–966, DOI: [10.1109/JSEN.2020.2968653](https://doi.org/10.1109/JSEN.2020.2968653) (2020).
18. Jabbooree, A. I., Khanli, L. M., Salehpour, P. & Pourbahrami, S. A novel facial expression recognition algorithm using geometry β -skeleton in fusion based on deep cnn. *Image Vis. Comput.* **134**, 104677 (2023).
19. Gupta, A., Hinchey, M. & Zalevsky, Z. Artificial intelligence and its applications. .
20. Sun, B. & et al. Multimodal attention networks for affective computing. *IEEE Transactions on Affect. Comput.* DOI: [10.1109/TAFFC.2020.2972216](https://doi.org/10.1109/TAFFC.2020.2972216) (2020).
21. Tripathi, S. & et al. Cross-modal transformers for multimodal affective state estimation. *Neural Networks* **144**, 439–450, DOI: [10.1016/j.neunet.2021.08.018](https://doi.org/10.1016/j.neunet.2021.08.018) (2021).
22. Wang, H. & et al. Missing-modality robust multimodal affect recognition via adaptive fusion. *Pattern Recognit.* **142**, 109677, DOI: [10.1016/j.patcog.2023.109677](https://doi.org/10.1016/j.patcog.2023.109677) (2023).
23. Sloane, G. *Staying gutsy during aging: the cost of stress on attention*. Ph.D. thesis, University of Essex (2024).
24. Priego-Torres, B. *et al.* Multimodal fusion of chest x-rays and blood biomarkers for automated silicosis staging. *J. Clin. Medicine* **14**, 8074 (2025).
25. Lokhande, H. & Ganorkar, S. R. Object detection in video surveillance using mobilenetv2 on resource-constrained low-power edge devices. *Bull. Electr. Eng. Informatics* **14**, 357–365 (2025).
26. Ali, H., Shifa, N., Benlamri, R., Farooque, A. A. & Yaqub, R. A fine tuned efficientnet-b0 convolutional neural network for accurate and efficient classification of apple leaf diseases. *Sci. Reports* **15**, 25732 (2025).
27. Li, Y., Huang, W.-C. & Song, P.-H. A face image classification method of autistic children based on the two-phase transfer learning. *Front. Psychol.* **14**, 1226470 (2023).
28. Ghiasi, M. M. & Zendehboudi, S. Application of decision tree-based ensemble learning in the classification of breast cancer. *Comput. biology medicine* **128**, 104089 (2021).
29. Almeida, J., Costa, M. & Silva, A. Ecg and gsr-based stress detection using lstm and cnn. *IEEE Transactions on Biomed. Eng.* **68**, 1227–1235, DOI: [10.1109/TBME.2021.3057221](https://doi.org/10.1109/TBME.2021.3057221) (2021).
30. Kumar, A., Sharma, P. & Singh, R. Multimodal stress detection using video and audio with 3d-cnn. *J. Affect. Comput.* **13**, 145–157, DOI: [10.1109/TAFFC.2022.3051371](https://doi.org/10.1109/TAFFC.2022.3051371) (2022).
31. Mao, L., Zhang, L. & Wang, Y. Ppg and respiratory rate based stress detection using random forest. *IEEE Access* **11**, 10456–10466, DOI: [10.1109/ACCESS.2023.3234087](https://doi.org/10.1109/ACCESS.2023.3234087) (2023).

32. Liu, Y., Zhang, J. & Wang, F. Stress detection using eye-tracking and facial expressions with rnn and svm. *Neural Comput. Appl.* **36**, 8395–8404, DOI: [10.1007/s00542-023-07902-5](https://doi.org/10.1007/s00542-023-07902-5) (2024).
33. Zhang, W., Li, H. & Song, Y. Speech and ecg-based stress detection using deep learning and cnn. *IEEE Transactions on Affect. Comput.* **14**, 340–352, DOI: [10.1109/TAFFC.2023.3134687](https://doi.org/10.1109/TAFFC.2023.3134687) (2023).
34. Lee, J., Kim, M. & Zhang, Y. Eeg and eye-tracking based stress detection using transformer and cnn. *IEEE Transactions on Affect. Comput.* **15**, 10–23, DOI: [10.1109/TAFFC.2024.3134532](https://doi.org/10.1109/TAFFC.2024.3134532) (2025).
35. Wang, Z., Liu, X. & Zhao, R. Stress detection using video and audio data with bilstm. *Neural Networks* **155**, 114–124, DOI: [10.1016/j.neunet.2024.12.022](https://doi.org/10.1016/j.neunet.2024.12.022) (2025).